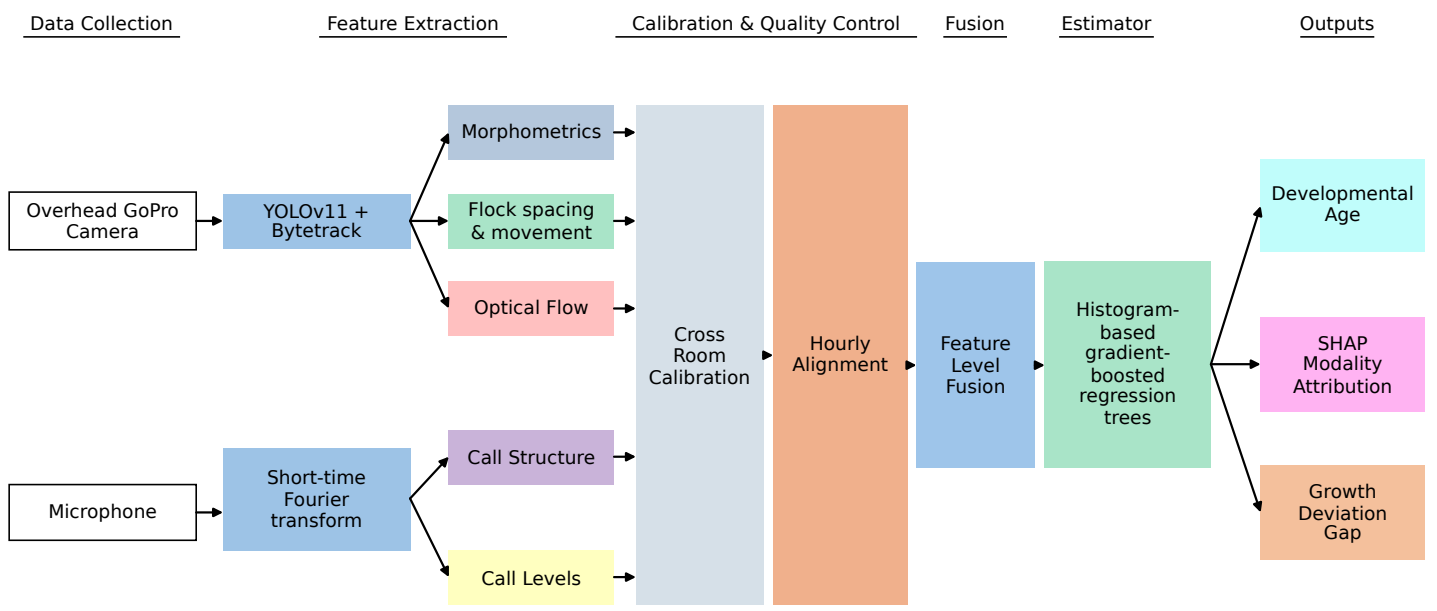
Data Collection
Feature Extraction
Calibration & Quality Control
Fusion
Estimator
Outputs
Overhead GoPro Camera
YOLOv11 + Bytetrack
Morphometrics
Flock spacing & movement
Optical Flow
Microphone
Short-time Fourier transform
Call Structure
Call Levels
Cross Room Calibration
Hourly Alignment
Feature Level Fusion
Histogram-based gradient-boosted regression trees
Developmental Age
SHAP Modality Attribution
Growth Deviation Gap